

Dominique Makowski

ASSISTANT PROFESSOR IN PSYCHOLOGY

University of Sussex, Brighton, UK

✉ D.Makowski@sussex.ac.uk | 🌐 realitybendinglab.com | 📧 [DominiqueMakowski](https://www.instagram.com/DominiqueMakowski) | 🐦 [Dom_Makowski](https://twitter.com/Dom_Makowski)



*Trained as a clinical neuropsychologist and CBT psychotherapist, I am an assistant professor at the University of Sussex, where I head the [Reality Bending Lab](http://realitybendinglab.com). My research has three strands. **Reality and its distortions:** visual illusions, deception and lie detection, misinformation and fake news, beliefs about AI-generated content, fiction, and altered states of consciousness. **The body in cognition:** how interoception, physiological states and emotion regulation shape what we take to be real. **Methods:** Bayesian statistics, computational models and psychophysiology, as well as the development of new software, measures and paradigms, released as open source (easystats, NeuroKit, cogmod) to support open and rigorous science.*

Experience



Assistant Professor (Current Position)

SCHOOL OF PSYCHOLOGY, UNIVERSITY OF SUSSEX

UK

2023 - Present

- Full teaching service on courses including Bayesian Statistics and Psychophysiology.
- Head of the [Reality Bending Lab](http://realitybendinglab.com).
- School-level leadership responsibilities, including research ethics, the research participation scheme and postgraduate programme development.



Research Fellow

CLINICAL BRAIN LAB, NANYANG TECHNOLOGICAL UNIVERSITY (NTU)

Singapore

2019 - 22

- In 2021, I got awarded the NTU "presidential postdoc fellowship", a competitive two-years grant meant as an intermediate step between postdoc and faculty. While integrated within a larger lab (the [Clinical Brain Lab](http://clinicalbrainlab.com), directed by [Pr Annabel Chen](#), I started my own research track on the neurocognitive and affective processes underlying beliefs about reality, managing my team (3 research assistants) and grant funding.
- In 2019, started my first research fellow position, working on a project funded by the ministry of defence about the neurocognitive correlates of deception. Reference: [Pr Annabel Chen](#).

Education



Doctor of Philosophy (PhD) - Psychology

UNIVERSITÉ PARIS CITÉ

France

2014 - 18



Certified Psychotherapist - Cognitive Behavioural Therapy (CBT)

FRENCH ASSOCIATION FOR COGNITIVE BEHAVIOURAL THERAPY (AFTCC)

France

2014 - 17



BSc, MSc - Clinical Neuropsychology and Cognitive Psychology

UNIVERSITÉ PARIS CITÉ

France

2009 - 14

Teaching

Module convenor (Bayesian Statistics)

UNIVERSITY OF SUSSEX

Brighton, UK

2024 - Present

- I teach the Bayesian framework and Bayesian modelling with brms (Master and PhD level).

Module teacher (Statistics)

UNIVERSITY OF SUSSEX

Brighton, UK

2023 - Present

- I teach lectures and practicals on statistics with R (Bachelor and Master level).

Module teacher (Psychophysiology)

UNIVERSITY OF SUSSEX

Brighton, UK

2023 - Present

- I teach lectures on Psychophysiology and Interoception in Cognitive Neuroscience modules (Bachelor and Master level).

Assistant Lecturer (Clinical Neuropsychology)

UNIVERSITÉ PARIS CITÉ

Paris, France

2017 - 18

- Modules (192 h; bachelor & master students): Neurology and psychiatry (main neurodegenerative, neurological and psychiatric disorders), assessment (examination, diagnostic, psychometry), rehabilitation and psychotherapy (CBT, mindfulness-based therapies, cognitive rehabilitation, practice supervision), general neuroscience and psychology (neuroanatomy, emotion regulation...). Reference: [Pr Isabelle Jambaque](#).

Teaching Assistant (Differential Cognitive Psychology)

Paris, France

UNIVERSITÉ PARIS CITÉ

2015 - 17

- Modules (144 h; 1st & 2nd year students): Experimental methodology (assessment and related problems, scientific reasoning, psychometric properties of tests and questionnaires, statistics...) and introduction to concepts such as intelligence, personality and creativity. Reference: Dr Christophe Mouchiroud.

Teaching Assistant (Cognitive Psychology)

Paris, France

UNIVERSITÉ PARIS CITÉ

2014 - 16

- Modules (48 h; 3rd year students): Models of working memory, mental imagery and categorisation. Reference: [Pr Todd Lubart](#).

CONSULTANCY AND KNOWLEDGE EXCHANGE

Instructor

Contracted Instructor

UK

SOUTH EAST DOCTORAL TRAINING ARC (SEDARC)

2026

- Recording of a re-usable *Introduction to GitHub* lecture for PhD students across the SEDarc partnership of six UK universities.

Invited Instructor (Psychophysiology)

Basel, Switzerland

UNIVERSITY OF BASEL

2024

- Full-day workshop (8 h; PhD students): Recording, Analyzing and Using Physiological Signals in Psychology. Reference: [Dr Dorothee Bentz](#).

Invited Instructor (Bayesian Statistics)

Paris, France

UNIVERSITÉ PARIS CITÉ

2024

- Workshop on Bayesian statistics for the members of a research lab.

Invited Instructor (Educational Course)

Montréal, Canada

ORGANIZATION FOR HUMAN BRAIN MAPPING (OHBM)

2023

- OHBM 2023 Educational Course: *Physiological Signals Provide Insights into Cognition*.

Invited Instructor (Programming and Data Science)

Zürich, Switzerland

UNIVERSITY OF ZÜRICH

2022

- Online workshop (24 h; PhD students): Advanced Python programming for signal processing and data analysis. Reference: [Dr Maike Krannich](#).

Invited Instructor (Mixed Models)

Singapore

NANYANG TECHNOLOGICAL UNIVERSITY (NTU)

2021

- Workshop on mixed models in R.

Developer

Contracted Developer

INSTITUTE FOR ADVANCED CONSCIOUSNESS STUDIES

2026

- Development of hyperscanning capabilities for an open-source portable EEG software.

Contract Research

Scientific Advisor

Singapore

AI STARTUP (CLIENT CONFIDENTIAL)

2025 - Present

- Scientific advising on an applied project on deception detection.

Contract Research

FRAGRANCE INDUSTRY (CLIENT CONFIDENTIAL)

2026

- Contract research on the relationship between smells and emotions.

CLINICAL

Assistant Neuropsychologist

Paris, France

DEPARTMENT OF ADULT PSYCHIATRY, PITIÉ-SALPÊTRIÈRE HOSPITAL

2013 - 14

- Neuropsychology internship (512 h) under [Dr Séverine Hatif](#). Assessment and diagnostic of psychiatric or neurological illnesses among adults, as well as development and application of cognitive rehabilitation programs.

Assistant Neuropsychologist

Paris, France

DEPARTMENT OF GERIATRIC PSYCHIATRY, SAINTE-ANNE HOSPITAL

2012 - 13

- Neuropsychology internship (304 h) under Emmanuel Guiliano. Assessment and diagnostic of psychiatric and neurological illnesses among an elderly population.

Assistant Neuropsychologist

Versailles, France

DEPARTMENT OF GERONTOLOGY, LA PORTE VERTE HOSPITAL

2012

- Neuropsychology internship (96 h) under Sophie Fleury. Observation of assessment and diagnostic of memory related deficits such as neurodegenerative diseases, somatic ageing-related illnesses or depression.

Software and Tools

I develop and maintain open-source software and measurement tools used across psychological and neuroscientific research.

Software		Measure	
easystats	R packages for accessible statistical modelling (bayestestR, parameters, performance, modelbased, ...)	Illusion Game	Parametric visual illusion task measuring illusion sensitivity
NeuroKit	Python toolbox for neurophysiological signal processing	Lie Scale	Self-report measure of lying styles and attitudes
Pyllusion	Python framework to generate visual illusions	MINT Scale	Multimodal interoception questionnaire
cogmod	R package for Bayesian cognitive models of decision-making tasks		
Neuropsychydia	Python package for creating experiments		

LANGUAGE

French



Native

Polish



Native

English



Fluent

Italian



B1 (learning)

TECHNIQUES

- **Bayesian Statistics:** Predictive and structural modelling within a Bayesian framework (Stan, brms), including custom distributional families and computational cognitive models.
- **Psychophysiology & EEG:** Acquisition, preprocessing and analysis of EEG and physiological signals (ECG, EDA, HRV).
- **Computational Modelling:** Application of ML/deep learning and computational modelling to psychological and neuroscientific data.
- **Open Science:** Strong commitment to reproducible research practices, open data/code sharing, and sustained contribution to open-source software communities.

Additional Training

TEACHING

Fellow of the Higher Education Academy (FHEA)

UK

ADVANCE HE (ATTAINMENT AGAINST THE PROFESSIONAL STANDARDS FRAMEWORK FOR TEACHING AND LEARNING SUPPORT IN HIGHER EDUCATION; ASSOCIATE FELLOW, AFHEA, IN 2024)

2025

CITI Certification

Online

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (ETHICS AND SCIENTIFIC INTEGRITY IN RESEARCH; BASIC COURSE AND REFRESHER)

2019, 2022

Research Integrity Certification

Singapore

NANYANG TECHNOLOGICAL UNIVERSITY (EPIGEUM RESEARCH INTEGRITY COURSE, SOCIAL & BEHAVIOURAL SCIENCES TRACK)

2019

Academic Teaching Certificate

France

UNIVERSITÉ OF PARIS (40H TRAINING IN PEDAGOGY AND INNOVATIVE TEACHING)

2017

CLINICAL

Certified Hypnotherapist

Singapore

NATIONAL GUILD OF HYPNOTISTS (100H - NOTE THAT ALTHOUGH TRAINED AND QUALIFIED, I DO NOT ENDORSE HYPNOSIS)

2019

Mindfulness based on Compassion and Insight (MBCI)

France

INSTITUTE FOR PSYCHIATRY, SAINTE-ANNE HOSPITAL (15H MASTER CLASS)

2015

First Aid Certificate

France

UNIVERSITÉ PARIS CITÉ (16H - LEVEL 1)

2016

NEUROIMAGING

Structural, Functional and Radiological Neuroanatomy

Online

NEUROCOURSES UK (32H)

2020 - 21

MRI Connectivity Analysis

Marseille, France

NATIONAL INSTITUTE OF NUCLEAR SCIENCES & TECHNIQUES - INSTN (32H)

2015

Functional MRI

Saclay, France

NATIONAL INSTITUTE OF NUCLEAR SCIENCES & TECHNIQUES - INSTN (28H)

2015

fMRI : Preprocessing and Analysis

Paris, France

SAINTE-ANNE HOSPITAL (32H)

2014 - 15

Signal Processing Theory and Application for EEG

Paris, France

ÉCOLE SUPÉRIEURE DE PHYSIQUE ET DE CHIMIE INDUSTRIELLE - ESPCI (24H)

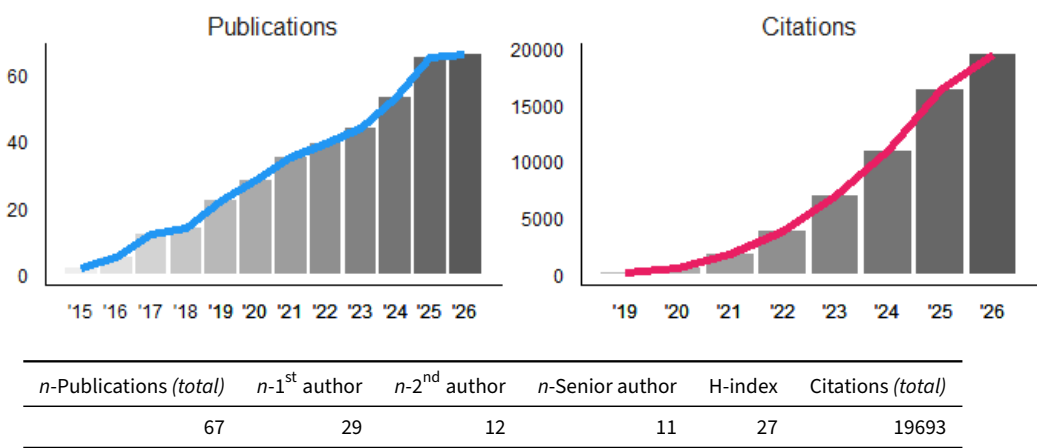
2014

STATISTICS AND PROGRAMMING

Creating Online Experiments in PsychoPy and Pavlovia	Online
UNIVERSITY OF NOTTINGHAM (24H)	2020
Scientific Programming with Python and Software Engineering Best Practices	Paris, France
TELECOM PARISTECH (16H)	2016
P4N: Python for Neuroscience (and Psychology)	Nottingham, UK
UNIVERSITY OF NOTTINGHAM (24H)	2016
Statistics and R for the Life Sciences (PH525.1x)	Online
HARVARDX, HARVARD UNIVERSITY (VERIFIED CERTIFICATE, VIA EDX)	2015
Machine Learning and Brain-to-Computer Interfaces	Paris, France
ÉCOLE SUPÉRIEURE DE PHYSIQUE ET DE CHIMIE INDUSTRIELLE - ESPCI (16H)	2014
MISCELLANEOUS	
History of Arts	Paris, France
ÉCOLE DU LOUVRE (74H)	2017– 18

Research Track Record

SCIENTIFIC IMPACT



GRANTS

Centre for Open Science: Funding Consciousness Research with Registered Reports (PI)	2024 - 26
CONSCIOUSNESS REGISTERED REPORTS: ILLUSION SENSITIVITY AND PHENOMENOLOGICAL CONTROL	\$28,163
TL@NTU’s Behavioural and Communication Science Programme (BCSP) Grant (co-PI)	2022 - 25
COGNITIVE, NEURAL AND SOCIAL FACTORS INVOLVED IN THE PERCEPTION OF REAL VS. FAKE INFORMATION	\$915,427
Singapore National Research Foundation’s Intra-CREATE Grant (co-PI)	2021 - 22
A NEUROSCIENTIFIC INVESTIGATION OF THE INTERACTION BETWEEN CROWDEDNESS AND ENVIRONMENT TYPOLOGY (NICE)	\$249,730
NTU Presidential Postdoctoral Fellowship (PI)	2020 - 22
’THAT’S FAKE NEWS’: MODULATING THE NEUROCOGNITIVE PROCESSES UNDERLYING THE BELIEF IN DECEPTIVE INFORMATION	\$360,000

AWARDS

Sussex School of Psychology 2024 Award Winner	2024
OUTSTANDING STUDENT SUPERVISION	
Commendation (Society for the Improvement of Psychological Science - SIPS)	2024
OBTAINED FOR THE NEUROKIT PYTHON PACKAGE	
University of Sussex Research Culture 2024 Awards Winner	2024
FOR ENGAGEMENT IN OPEN SCIENCE	£1,000
Mission Award (Society for the Improvement of Psychological Science - SIPS)	2023
OBTAINED FOR THE EASYSTATS COLLECTION OF R PACKAGES	

Best Article Award (Psychonomic Society)

OBTAINED FOR [MAKOWSKI ET AL. \(2019\)](#)

2020

\$1,000

Commendation (Society for the Improvement of Psychological Science - SIPS)

OBTAINED FOR THE BAYESTEST R PACKAGE ([MAKOWSKI ET AL., 2019](#))

2020

STUDENT AWARDS

My students are also consistently winning awards for their work in my lab.

SAGE Publications Prize for Outstanding Application of Statistical Methods

AWARDED TO EMMA BENN (3RD YEAR DISSERTATION)

2026

Junior Research Assistant (JRA)

SCHOLARSHIP AWARDED TO OLIVER COLLINS

2026

SoCoBio Summer Internship

SCHOLARSHIP AWARDED TO EVA MALA

2025

Sussex Partnership NHS Foundation Trust Prize

AWARDED TO SAM MARINE (3RD YEAR DISSERTATION) FOR HIGHEST PROJECT MARK

2025

Adam Weiler Impact Award

AWARDED TO ROB DICKINSON (PHD STUDENT)

2024

SoCoBio Summer Internship

SCHOLARSHIP AWARDED TO GREGORY HEANEY

2024

Junior Research Assistant (JRA)

SCHOLARSHIP AWARDED TO JING XU

2023

Junior Research Assistant (JRA)

SCHOLARSHIP AWARDED TO AUZ MOORE

2023

EDITORIAL RESPONSIBILITIES

Statistical Editor

ROYAL SOCIETY OPEN SCIENCE

2025 - present

Associate Editor

FRONTIERS IN PSYCHOLOGY: *Quantitative Psychology and Measurement*

2020 - 2024

Regular Reviewer

JOURNAL OF OPEN SOURCE SOFTWARE

2018 - present

Reviewer

BIOLOGICAL PSYCHIATRY, PERSONALITY AND INDIVIDUAL DIFFERENCES, SCIENTIFIC REPORTS, ACTA PSYCHOLOGICA, ...

2017 - present

Teaching Record

ACADEMIC

I have given academic lectures (total hours since 2014: **424**) on the following topics:

- **Bayesian Statistics**
 - Bayesian framework
 - Bayesian models with the *brms* R package
- **Signal Processing**
 - Bodily signals acquisition and preprocessing
 - Heart rate
 - Electrodermal activity
- **Psychophysiology and Interoception**
 - Embodied Cognition
 - Interoceptive Models
 - Psychophysiology
 - Heart rate variability
- **Neuropsychology** (192 hours; 2nd year, 3rd year and master level)
 - Clinical Neuropsychology (Neuropsychological syndromes; Neuropsychological examination and diagnostic; Cognitive rehabilitation; Clinical practice supervision; Neuropsychology in psychiatry)

- Cognitive Neuroscience (Neuroanatomy; Emotion Regulation; Mindfulness; Memory; Neuroimaging)
- **Differential Psychology** (144 hours; 1st year level)
 - Hands-on Lectures (Experimental Methodology; Scientific Reasoning; Psychometric properties of tests and questionnaires; Introductory Statistics)
 - Theory (Intelligence; Personality; Creativity)
- **Cognitive Psychology** (48 hours; 3rd year level)
 - Working memory
 - Mental imagery and categorisation
- **Programming for Psychologists** (24 hours)
 - Python Programming
 - Experiment creation using Python
- **Open Science and Research Tools** (MRes level workshops)
 - Version control and collaboration with Git and GitHub
 - Agentic coding tools for research
- **Other** (16 hours): Internet; Research Ethics; Digital Rights

CURRICULUM DEVELOPMENT

- **MSc Advanced Computational Psychology** (2026 - Present). Development of a new postgraduate programme at the University of Sussex, covering its curriculum structure, module content and learning outcomes.
- **Bayesian Statistics** (2024 - Present). Designed a new module from scratch, taught at Master and PhD level.

SUPERVISION

- **Research staff:** mentored more than 7 full-time research assistants, several of whom moved on to funded PhD positions.
- **Doctoral students:** PhD co-supervision since 2019 (NTU, Singapore) and PhD supervision at Sussex (2023 - Present).
- **Dissertation and placement students:** each year at Sussex I supervise around 9 undergraduate dissertation students, 3 master's dissertation students and 2 placement students working full time in the lab, as well as funded summer interns (Junior Research Associate and SoCoBio schemes). See *Student Awards* above.

EXAMINING

- **Internal examiner for four PhD theses** at the University of Sussex (2024 - Present), on consciousness science, interoception and quantitative methods (candidates supervised by Pr Anil Seth, Pr Hugo Critchley and Pr Andy Field).
- Examiner and moderator for undergraduate and master's dissertations (2023 - Present).

Service and Leadership

INSTITUTIONAL SERVICE

School of Psychology, University of Sussex

ACADEMIC LEAD, RESEARCH PARTICIPATION SCHEME (SONA)

2025 - Present

- Redesigned the credit allocation and study approval rules of the School's research participation scheme to make it fairer and more transparent for both students and researchers.
- Trialled changes aimed at increasing the educational value of participation for students while simultaneously improving data quality, including measures to address low-effort responding.

School of Psychology, University of Sussex

PROGRAMME DEVELOPER, MSc ADVANCED COMPUTATIONAL PSYCHOLOGY

2026 - Present

- Development of a new postgraduate programme, including its curriculum structure, module content and learning outcomes.

University of Sussex

MEMBER, CROSS-SCHOOLS RESEARCH ETHICS COMMITTEE (C-REC)

2023 - Present

OPEN SCIENCE AND RESEARCH CULTURE

Centre for Open Science and Research Reform

University of Sussex

ORGANIZER, MORSE SEMINAR SERIES

2024 - Present

- Member of the Centre, where I convene MORSE (Methods for Open and Reproducible Science), a seminar series on openness, reproducibility and replicability in psychological science.

British Neuroscience Association (BNA) Members' Meeting

OPEN SCIENCE CHAMPION (INVITED DEBATE)

2024

- Invited to argue the open science position in the meeting's interactive debate, *A cultural revolution in sustainable neuroscience is needed more than a cultural revolution in open neuroscience now*, alongside Dr Charlotte Rae (Sustainability Champion) and Hannah Hope (Wellcome).

CONFERENCE ORGANIZATION

ESCAN 2026, Sapienza University, Rome, Italy

CO-ORGANIZER, Artificial Emotion SYMPOSIUM

2026

- Artificial Emotion: Affective Responses and Interaction with Artificial Intelligence*: a four-talk symposium on affective, physiological and neural responses to artificial intelligence, co-organised with Marco Sperduti (Università di Roma Tor Vergata) for the meeting of the European Society for Cognitive and Affective Neuroscience.

EARLIER SERVICE

ED261-3CH doctoral school

*Representation of the PhD students
at the university level*

ELECTED REPRESENTATIVE

2016 - 18

Memory & Cognition Lab

*Organization of seminars, local
conferences and social events*

COMMUNITY MANAGER

2014 - 15

Student Association (BDE)

*Organization of social events for
students*

ACTIVE MEMBER

2009 - 12

Publications

REALITY PERCEPTION AND MANIPULATION

Makowski, D., & Neves, A. (*under review*). [Testing the relationship between phenomenological control and illusion sensitivity](#). *PsyArXiv*. (Stage 2 Registered Report)

Dickinson, R., **Makowski, D.**, van Marwijk, H., & Ford, E. (*under review*). [Bad News: Testing gamified inoculation against misinformation](#). *PsyArXiv*.

Makowski, D., Te, A. S., Neves, A., Kirk, S., Liang, N. Z., Mavros, P., & Chen, S. A. (2025). [Too beautiful to be fake: Attractive faces are less likely to be judged as artificially generated](#). *Acta Psychologica*, 252, 104670.

Dickinson, R., **Makowski, D.**, van Marwijk, H., & Ford, E. (2025). [Interventions for combating COVID-19 misinformation: A systematic realist review](#). *PLOS One*, 20(4), e0317592.

Dickinson, R., **Makowski, D.**, van Marwijk, H., & Ford, E. (2024). [Exploring the Role of News Outlets in the Rise of a Conspiracy Theory: Hydroxychloroquine in the Early Days of COVID-19](#). *COVID*, 4(12), 1873-1896.

Makowski, D., Lau, Z. J., Pham, T., Te, A. S., Kirk, S., Liauw, C., & Chen, S. A. (2024). [The heart can lie: a preliminary investigation of the role of interoception and theory of mind in deception](#). *Current Psychology*, 1-12.

Makowski, D., Te, A. S., Kirk, S., Liang, N. Z., & Chen, S. A. (2023). [A novel visual illusion paradigm provides evidence for a general factor of illusion sensitivity and personality correlates](#). *Scientific Reports*, 13(1), 6594.

Makowski, D., Lau, Z. J., Pham, T., Boyce, W. P. & Chen, S. A. (2021). [A Parametric Framework to Generate Visual Illusions using Python](#). *Perception*, 50(11), 950-965.

Makowski, D., Pham, T., Lau, Z. J., Raine, A., & Chen, S. H. (2021). [The Structure of Deception: Validation of the Lying Profile Questionnaire](#). *Current Psychology*, 1-16.

Makowski, D., Sperduti, M., Pelletier, J., Blondé, P., La Corte, V., Arcangeli, M., Zalla, T., Lemaire, S., Dokic, J., Nicolas, S., & Piolino, P. (2019). [Phenomenal, Bodily and Brain Correlates of Fictional Reappraisal as an Implicit Emotion Regulation Strategy](#). *Cognitive, Affective, & Behavioral Neuroscience*, 19(4), 1-21.

Makowski, D.*, Sperduti, M.*, Nicolas, S. & Piolino, P. (2017). ["Being there" and remembering it: Presence improves Memory Encoding](#). *Consciousness & Cognition*, 53, 194-202.

Sperduti, M.*, **Makowski, D.***, Arcangeli, M., Wantzen, P., Zalla, T., Lemaire, S., Dokic, J., Pelletier, J. & Piolino, P. (2017). [The Distinctive Role of Executive Functions in Implicit Emotion Regulation](#). *Acta Psychologica*, 173, 13-20.

Tuena, C., Serino, S., Gaston-Bellegarde, A., Orriols, E., **Makowski, D.**, Riva, G. & Piolino, P. (2017). [How Virtual Embodiment Affects Episodic Memory Functioning: a Proof-of-Concept Study](#), *Annual Review of Cybertherapy and Telemedicine*, 15, 93-107.

Sperduti, M.*, Arcangeli, M.*, **Makowski, D.**, Wantzen, P., Zalla, T., Lemaire, S., Dokic, J., Pelletier, J. & Piolino, P. (2016). [The paradox of fiction: Emotional response toward fiction and the modulatory role of self-relevance](#). *Acta psychologica*, 165, 53-59.

INTEROCEPTION, EMOTION REGULATION, NEUROPHYSIOLOGY

Makowski, D., Neves, A., Benn, E., Bennett, M., & Poerio, G. (*under review*). [The Mint Scale: A fresh validation of the multimodal interoception questionnaire and comparison to the MAIA, BPQ and IAS](#). *PsyArXiv*.

Neves, A., Pfaff, M., Scharte, R. E., Nogueira-Arjona, R., Poerio, G., & **Makowski, D.** (under review). [Mega-analysis of the Interoceptive Accuracy Scale \(IAS\) structure and its dispositional correlates](#). *PsyArXiv*.

Scharte, R. E., Greenwood, K., Romero Sanchiz, P., Neves, A., **Makowski, D.**, & Nogueira-Arjona, R. (under review). [Interoception and substance use: A systematic review and meta-analysis](#). *PsyArXiv*.

Makowski, D. (under review). [Towards an active inference personality framework: The Deep-Self Predictive Cascade model](#). *PsyArXiv*.

Pham, T., Johnco, C. J., Lau, Z. J., **Makowski, D.**, & Forbes, M. K.* (2025). [Which heart rate variability \(HRV\) indices should I use for psychophysiological research? A data-driven approach to identifying clusters of HRV indices](#). *Psychophysiology*, 62(10), e70164. (*co-last authors)

Makowski, D., Te, A. S., Neves, A., & Chen, S. H. A. (2025). [Measuring depression and anxiety with 4 items? Adaptation of the PHQ-4 to increase its sensitivity to subclinical variability](#). *Collabra: Psychology*, 11(1), 143431.

Lau, Z. J.*, Pham, T.*, Chen, S. A., & **Makowski, D.** (2022). [Brain Entropy, Fractal Dimensions and Predictability: A Review of Complexity Measures for EEG in Healthy and Neuropsychiatric Populations](#). *European Journal of Neuroscience*, 56(7), 5047-5069.

Makowski, D., Te, A. S., Pham, T., Lau, Z. J., & Chen, S. H. (2022). [The Structure of Chaos: An Empirical Comparison of Fractal Physiology Complexity Indices Using Neurokit2](#). *Entropy*, 24(8), 1036.

Pham, T., Lau, Z. J., Chen, S. H. A., & **Makowski, D.** (2021). [Heart Rate Variability in Psychology: A Review of HRV Indices and an Analysis Tutorial](#). *Sensors*, 21(12), 3998.

Makowski, D., Pham, T., Lau, Z. J., Brammer, J. C., Lespinasse, F., Pham, H., Schötzel, C., & Chen, S. A. (2021). [NeuroKit2: A Python Toolbox for Neurophysiological Signal Processing](#). *Behavior Research Methods*, 1-8.

Makowski, D., Sperduti, M., Blondé, P., Nicolas, S., & Piolino, P. (2019). [The heart of cognitive control: Cardiac phase modulates processing speed and inhibition](#). *Psychophysiology*, 57(3).

Makowski, D., Sperduti, M., Lavallée, S., Nicolas, S., & Piolino, P. (2019). [Adaptation and Validation of a Short French Version of the Affective Style Questionnaire](#). *International Journal of Cognitive Therapy*, 13(2), 146-158.

Makowski, D.*, Sperduti, M.*, Lavallée, S., Nicolas, S. & Piolino, P. (2019). [Dispositional mindfulness attenuates the emotional attentional blink](#). *Consciousness & Cognition*, 67, 16-25.

Makowski, D., Sperduti, M., Blanchet, S., Nicolas, S. & Piolino, P. (2015). [Régulation Emotionnelle face au Déclin Cognitif dans le Vieillessement : Un Faux Paradoxe ? | Emotion regulation and the cognitive decline in aging: beyond the paradox](#). *Gériatrie et Psychologie Neuropsychiatrie du Vieillessement*, 13(3), 301-8.

STATISTICS

Makowski, D., Neves, A., & Field, A. P. (under review). [Introducing the Choice-Confidence \(CHOCO\) model for bimodal data from subjective ratings: Application to the effect of attractiveness on reality beliefs about AI-generated faces](#). *PsyArXiv*.

Makowski, D. (under review). [The cogmod R package: Easy-to-use Bayesian cognitive models for reaction times, choices and subjective ratings, with applications to computational neuropsychology](#). *PsyArXiv*.

Geller, J., Angele, B., & **Makowski, D.** (under review). [This is not the ex-Gaussian model you are looking for: On the default parameterization of Bayesian ex-Gaussian models in brms](#). *PsyArXiv*.

Lüdecke, D., Makowski, A. C., Klein, J., Ben-Shachar, M. S., & **Makowski, D.** (2026). [Choosing informative priors in Bayesian regression models: A simulation study and tutorial using Stan and R](#). *Frontiers in Psychology*, 17, 1856582.

Fernandez, K., **Makowski, D.**, & Fisher, C. (2025). [SequentialSamplingModels.jl: Simulating and evaluating cognitive models of response times in Julia](#). *JuliaCon Proceedings*, 7(78), 186.

Makowski, D., & Waggoner, P. D. (2023). [Where are we going with statistical computing? From mathematical statistics to collaborative data science](#). *Mathematics*, 11(8), 1821.

Thériault, R., Ben-Shachar, M. S., Patil, I., Lüdecke, D., Wiernik, B. M., & **Makowski, D.** (2024). [Check your outliers! An introduction to identifying statistical outliers in R with easystats](#). *Behavior Research Methods*, 1-11.

Makowski, D., Ben-Shachar, M. S., Wiernik, B. M., Patil, I., Thériault, R., & Lüdecke, D. (2025). [modelbased: An R package to make the most out of your statistical models through marginal means, marginal effects, and model predictions](#). *Journal of Open Source Software*, 10(109), 7969.

Patil, I., **Makowski, D.**, Ben-Shachar, M. S., Wiernik, B. M., Bacher, E., & Lüdecke, D. (2022). [datawizard: An R Package for Easy Data Preparation and Statistical Transformations](#). *Journal of Open Source Software*, 7(78), 4684.

Lüdecke, D., Patil, I., Ben-Shachar, M., Wiernik, B., Waggoner, P., & **Makowski, D.** (2021). [see: An R Package for Visualizing Statistical Models](#). *Journal of Open Source Software*, 6(64), 3393.

Lüdecke, D., Ben-Shachar, M., Patil, I., Waggoner, P., & **Makowski, D.** (2021). [performance: An R Package for Assessment, Comparison and Testing of Statistical Models](#). *Journal of Open Source Software*, 6(60), 3139.

Ben-Shachar, M. S., Lüdecke, D., & **Makowski, D.** (2020). [effectsize: Estimation of Effect Size Indices and Standardized Parameters](#). *Journal of Open Source Software*, 5(56), 2815.

Lüdecke, D., Ben-Shachar, M. S., Patil, I., & **Makowski, D.** (2020). [parameters: Extracting, Computing and Exploring the Parameters of Statistical Models using R](#). *Journal of Open Source Software*, 5(53), 2445.

Makowski, D., Ben-Shachar, M. S., Patil, I., & Lüdecke, D. (2020). [Methods and Algorithms for Correlation Analysis in R](#). *Journal of Open Source Software*, 5(51), 2306.

Makowski, D., Ben-Shachar, M. S., Chen, S. H., & Lüdecke, D. (2019). [Indices of Effect Existence and Significance in the Bayesian Framework](#). *Frontiers in Psychology*, 10.

Makowski, D., Ben-Shachar, M., & Lüdecke, D. (2019). [bayestestR: Describing effects and their uncertainty, existence and significance within the Bayesian framework](#). *Journal of Open Source Software*, 4(40), 1541.

Lüdecke, D., Waggoner, P., & **Makowski, D.** (2019). [insight: A unified interface to access information from model objects in R](#). *Journal of Open Source Software*, 4(38), 1412.

Makowski, D. (2018). [The psycho Package: An Efficient and Publishing-Oriented Workflow for Psychological Science](#). *Journal of Open Source Software*, 3(22), 470.

MISCELLANEOUS

Geller, J., Gomez, P., Buchanan, E., & **Makowski, D.** (2025). [A distributional response time analysis of the perceptual disfluency effect](#). *Journal of Cognition*, 8(1), 60.

Rinaldi, L. J., Agbude, R., Andermane, N., **Makowski, D.**, Forster, S., Simner, J., & Ward, J. (2026). [Attention and physiological responses to task-irrelevant sounds in misophonia](#). *Journal of Clinical and Experimental Neuropsychology*, 48(3), 266-280.

Salgues, S., Jacquot, A., **Makowski, D.**, Tahar, C., Baekeland, J., Arcangeli, M., ... & Sperduti, M. (2024). [Self-reference and emotional reaction drive aesthetic judgment](#). *Scientific Reports*, 14(1), 19699.

Lee, H., Jacquot, A., **Makowski, D.**, Arcangeli, M., Dokic, J., Piolino, P., & Sperduti, M. (2024). [The beauty and the self: A common mnemonic advantage between aesthetic judgment and self-reference](#). *Psychology of Consciousness: Theory, Research, and Practice*, 11(1), 71.

Rose, L., Kovarski, K., Caetta, F., **Makowski, D.**, & Chokron, S. (2024). [Beyond empathy: Cognitive capabilities increase or curb altruism in middle childhood](#). *Journal of Experimental Child Psychology*, 239, 105810.

Lee, H., Jacquot, A., **Makowski, D.**, Arcangeli, M., Dokic, J., Piolino, P., & Sperduti, M. (2023). [Beauty is in the eye of the beholder: Evidence from a common mnemonic advantage between aesthetics judgement and self-reference](#). *Psychology of Consciousness: Theory, Research, and Practice*.

Kovarski, K., Dos Reis, J., Chevais, C., Hamel, A., **Makowski, D.**, & Sperduti, M. (2022). [Movie editing influences spectators' time perception](#). *Scientific Reports*, 12(1), 20084.

Blondé, P., Sperduti, M., **Makowski, D.**, & Piolino, P. (2022). [Bored, distracted, and forgetful: The impact of mind wandering and boredom on memory encoding](#). *Quarterly Journal of Experimental Psychology*, 75(1), 53-69.

Blondé, P., **Makowski D.**, Sperduti, M., & Piolino, P. (2020). [In Medio Stat Virtus: intermediate levels of mind wandering improve episodic memory encoding in a virtual environment](#). *Psychological Research*, 85(4), 1613-1625.

Girardeau, J. C., Blondé, P., **Makowski D.**, Abram, M., Piolino, P., & Sperduti, M. (2020). [The impact of state and dispositional mindfulness on prospective memory: A virtual reality study](#). *Consciousness and Cognition*, 81, 102920.

Makowski, D., Dutriaux, L. (2017). [Neuropsychia.py: A Python Module for Creating Experiments, Tasks and Questionnaires](#), *Journal of Open Source Software*, 2(19), 259.

Sperduti, M., Armougum, A., **Makowski, D.**, & Piolino, P. (2017). [Interaction between attentional systems and episodic memory encoding: the impact of conflict on binding of information](#), *Experimental Brain Research*, 235(12), 3553-3560.

Nicolas, S., & **Makowski, D.** (2017). [Centenaire Ribot \(I\). La réception de l'oeuvre de Théodule Ribot chez l'éditeur Ladrance \(1870-1873\)](#). *Bulletin de Psychologie*, 70(3), 163-178.

Sperduti, M., **Makowski, D.**, Blondé, P. & Piolino, P. (2017). [Meditation and successful aging: can meditative practices counteract age-related cognitive decline?](#), *Gériatrie et Psychologie Neuropsychiatrie du Vieillissement*, 15(2), 205-13.

Nicolas, S., & **Makowski, D.** (2016). [Can mental fatigue be measured by Weber's compass? Alfred Binet's answer on the value of aesthesiometry \(tactile sensitivity\) as an objective measure of mental fatigue](#). *European Yearbook of the History of Psychology*, 2, 11-46.

Sperduti, M., **Makowski, D.**, & Piolino, P. (2016). [The protective role of long-term meditation on the decline of the executive component of attention in aging: a preliminary cross-sectional study](#). *Aging, Neuropsychology, and Cognition*, 23(6), 691-702.

BOOK CHAPTERS

Mavros, P., Olszewska-Guizzo, A., & **Makowski, D.** (2023). [Disentangling the Social from the Aesthetic: Neuropsychology for Urban Design](#). In *Future Cities—City Futures: Emerging Urban Perspectives*. TU Delft OPEN Publishing.

Makowski, D., Chen, S. H. A., Larsen, R. R., Boyle, G. J., & Lilienfeld, S. O. (2023). [Clinical Neuropsychology in the Era of Neuroimaging](#). In *The SAGE Handbook of Clinical Neuropsychology*. SAGE Publications.

Communications

GENERAL AUDIENCE / MEDIA COVERAGE

- Research featured in online journal, [Attractiveness shapes beliefs about whether faces are real or AI-generated, study finds](#). 07 July 2025, psypost.org
- Invited on the "Learning Bayesian Statistics" Podcast, [#55 Neuropsychology, Illusions & Bending Reality, with Dominique Makowski](#). 01 Feb 2022, learnbayesstats.com
- Speaker at the Pint of Science, [Back to the Matrix : pilule bleue ou rouge ?](#). 16 May 2018, Paris, France.
- Invited to participate in a Public Debate, [Art & science discussion on sensory illusions](#). 24 November 2017, Bordeaux, France.
- Published article in magazine, [Neuroanatomie de la réalité : quand le réel n'est plus](#). 2016, Les yeux brouillés, 1, 48-55.
- Featured on National TV, [The effect of violent images on the brain and body](#). 2015, Testé sous contrôle médical, France 4
- Research Featured in Online journal, ["Avengers, l'ère d'Ultron" en 3D : aussitôt vu, aussitôt oublié ?](#) 2015, francetvinfo.fr

References

Do not hesitate to contact them of any inquiry.

- **Pr Annabel Chen** (Postdoc supervisor): AnnabelChen@ntu.edu.sg
- **Dr Marco Sperduti** (PhD mentor): marco.sperduti@parisdescartes.fr
- **Pr Pascale Piolino** (PhD cosupervisor): pascale.piolino@parisdescartes.fr
- **Pr Serge Nicolas** (PhD cosupervisor): serge.nicolas@parisdescartes.fr